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Department of Artificial Intelligence
Intelligent Multi-Agent and Expert Systems (AI401), Semester VII

Instructions

1. Save your file as LAB_No_Roll_No.doc (e.g. LAB_12_U21AI001.doc).
2. Paste a snapshot of each step followed along with the result(s).
3. Record your observations and comments after the results.

Experiment 12

Expert System Application — A Financial Advisor with Scoring Rules and Hard Constraints

Aim

To build an advisory expert system that converts a client profile into an asset allocation through scoring rules and overriding constraints, to explain each recommendation from the rules that produced it, and to evaluate the advice over simulated return scenarios rather than by inspection.

Theory

A financial advisor differs from the diagnostic system of Experiment 11 in what it is asked for: not the identity of a hidden state but a recommendation, which cannot be checked against a known right answer. Such systems are usually built in two layers. A scoring layer converts the client's circumstances — age, investment horizon, stability of income, dependants, stated tolerance for loss — into a single risk score, and bands of that score map to standard allocations between equity, bonds and cash. A constraint layer then sits above it with rules that override the score outright: high-interest debt is repaid before anything is invested, an emergency reserve is funded before risk is taken, and money needed within three years does not go into equity. The two layers are different kinds of knowledge, and mixing them into one score is the commonest fault in advisory systems, because a weighted sum lets a high score elsewhere buy away a condition that was meant to be absolute.

The recommendation cannot be graded against truth, so it is evaluated against the distribution of outcomes it implies. Drawing many return paths for each asset class over the client's horizon and compounding the allocation along each gives a distribution of final values, from which the median, the fifth percentile and the probability of ending below the amount invested can be read. The median says what the advice is for; the fifth percentile says what it risks. An allocation that raises the median while lowering the fifth percentile has not been shown to be better, and for a client who cannot absorb the bad case it is worse.

Algorithm

Algorithm 1 — Scoring rules, constraints and explanation

1. Define the client attributes and the points each contributes to the risk score, the score bands and the allocation of equity, bonds and cash attached to each band. Separately define the constraint rules that override an allocation, each with the condition that triggers it and the allocation or cap it imposes.
2. For a client, compute the score, select the band, apply any constraint that fires, and return the final allocation together with a trace naming the band and every constraint that acted. Run at least three contrasting clients and print the allocation and the trace for each.

Algorithm 2 — Evaluating the advice over scenarios

1. Give each asset class an annual return drawn from a normal distribution with a stated mean and standard deviation, and simulate at least a thousand paths over the client’s horizon, compounding the portfolio return of the recommended allocation along each path.
2. Report for every client the median final value, the fifth percentile and the probability of finishing below the amount invested, and repeat the evaluation for the allocation the scoring layer would have given had the constraints not been applied.

Procedure and Result

Paste the snapshots of the steps followed and the output produced by the program below, including the score, band, final allocation and explanation trace for each client, and the table of median, fifth percentile and probability of loss for the recommended allocations and for the unconstrained ones.

Observations

Comment on which client the constraint layer changed most and what the score alone would have advised, on what the fifth percentile adds that the median does not, on whether the explanation trace would satisfy a client who asked why they were told to hold cash, and on what the simulation does not capture about repaying debt before investing.

Exercises

1. Generate a population of clients and compare the system with and without the constraint layer. Report how many recommendations change, and for each constraint separately report the mean fifth-percentile outcome of the clients it affects under both versions. State which constraint does the most work and whether any of them could be folded into the score instead.
2. Examine the band boundaries. Report the recommended equity percentage for scores on either side of one boundary, and the largest change in equity that a single extra point of score can cause. Then replace the bands with an allocation interpolated smoothly between the band centres, report the same figures, and compare the two versions for a client sitting at the boundary over a ten-year horizon.